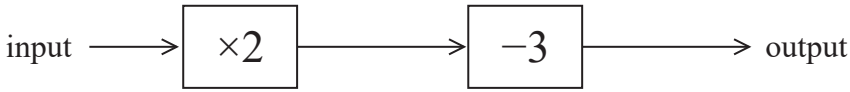
10 Here is a number machine.



(a) What is the **output** when the input is 4?

$4 \times 2 - 3 = 8 - 3 = 5.$

1 mark
Output5

.....

(1)

(b) What is the **input** when the output is 11?

Reverse the machine: add3, then divide by2.

1 mark
Inverse operations

$(11 + 3) \div 2 = 14 \div 2 = 7.$

1 mark
Input7

.....

(2)

(c) Show that there is an input for the machine for which the output is the same as the input.

$2x - 3 = x$

(+3) (+3)

$2x = x + 3$

(−x) (−x)

$x = 3$

1 mark
Set input = output

1 mark
Input and output3

Check: input3 gives $3 \times 2 - 3 = 3$.

(2)

(Total for Question 10 is 5 marks)